Sewon Kim

prepared7913@gmail.com
wontothree.github.io

Research Interests

Optimal Control and **Robot Learning**. I am interested in developing scalable learning and control methods and their theoretical foundations for robotics.

Education

University of Seoul Mar 2022 - Aug 2027
B.S. in Electrical and Computer Engineering, Minor in Mathematics Seoul, South Korea

Experience

Robotics and Physical AI Lab, Yonsei University Aug 2026 - Present
Intern (Advisor: Yonghyeon Lee) Seoul, South Korea

PDE and AI Lab, Yonsei University Sep 2024 - Present
Intern (Advisor: Dohyun Kwon) Seoul, South Korea

InBody AI Scale Team, InBody Jan 2024 - Jan 2025
Computer Vision Engineer Seoul, South Korea

Control and Dynamic System Lab, University of Seoul Mar 2023 - Dec 2024
Intern (Advisor: Gyunghoon Park) Seoul, South Korea

Projects

NAMUH X A1: Autonomous Air Purification Agent Aug 2025 - Sep 2025
4th Place (/200) ($2,200) — SK NAMUHX A1: Autonomous Navigation Path Optimization

Independently implemented full-stack autonomous air purifying agent in NumPy from scratch, covering MCL, MPPI, and TSP.

F1TENTH: Autonomous Racing with MPPI Mar 2024 - Dec 2024
5th Place (/12) — 22nd F1tenth Autonomous Grand Prix at CDC Milan 2024

Led the project with 2 undergrads and implemented a full-stack physical autonomous navigation system for racing from scratch, covering SLAM, MCL, Raceline Opt, MPPI.

Cart Pole System Design and Control with NMPC Jul 2024 – Sep 2024
3rd Place (/25) ($220) — UOS ECE Innovation Fair

Independently developed a physical Cart-Pole system for $272, covering modeling, motor control, sensor integration, state estimation, embedded systems, and MPC.

InBody AI Scale Jan 2024 - Jan 2025
Consumer Electronic Show 2025

Developed height estimation model (acc 99.4%) and face recognition system of "InBody AI Scale", automated body composition analyzer that reduces measurement time from 120s to 40s.

ECG data-driven cardiovascular diagnosis device Jul 2023 - Jul 2023
3rd Place (/12) ($1450) — InBody Hackathon 2023

Developed a fast prototype.

Autonomous mission mobile robot Jul 2023 - Jul 2023
3rd Place (/20) ($220) — Object Detection-Based Solar Tracking Robot Challenge, Korea University

Sidaeting Mar 2022 - Jun 2022
1,700 User

Implemented a matching algorithm that combinatorially optimizes collective happiness from users' ideal type data and developed the frontend for "Sidaeting", a dating service launched by UOSLIFE.

Skills

- Languages: Korean, English
- Programming Languages: C, C++, Java, JavaScript, Julia, Python, SQL, HTML, CSS
- Frameworks: Isaac Sim/Lab, ROS 1/2 (Melodic, Noetic, Foxy, Humble, Jazzy), MuJoCo, PyTorch

Services

Institute of Innovation for Future Army, Republic of Korea Army Jan 2025 - Jul 2026
Drone Researcher; Mandatory Military Service Daejeon, South Korea

AI Robotics KR Sep 2023 - Sep 2024
Management Team Seoul, South Korea

SW Educational Outreach Program 'Let's play software' Apr 2022 - Jan 2023
Software Elementary School Teacher Seoul, South Korea